Netflix Exploratory Data Analysis (EDA)

Project Overview

This project performs an **Exploratory Data Analysis (EDA)** on the Netflix dataset to uncover meaningful insights and patterns in Netflix's content library.

The analysis explores:

- Movies vs. TV Shows
- Content added over the years
- Content distribution by country
- Ratings and audience categories
- Popular genres and categories
- Movie duration and TV show seasons
- Trends in Netflix content over time

The goal is to **clean, transform, analyze, and visualize the data** to understand how Netflix's content library has evolved and identify key trends using **Python, Pandas, Matplotlib, and Seaborn**.

Objective

To use data-driven analysis and visualization techniques to extract **actionable insights from Netflix's content catalog** and better understand its content distribution, growth, ratings, genres, and release trends.

In [1]:

```
#Importing all the necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#from word cloud import WordCloud
```

In [2]:

```
df=pd.read_csv("netflix.csv")
df.head()
#any column having more than 50% unique values
#that column never contribute in project
```

Out[2]:

	show_id	type	title	director	cast	country	date_added	release_year	rating
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA

In [3]:
```
df.columns
```

Out[3]:
```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'dat
e_added',
       'release_year', 'rating', 'duration', 'listed_in', 'descriptio
n'],
      dtype='str')
```

In [4]:
```
df.rename(columns={'listed_in':'genre'},inplace=True)
df.columns
# Renaming the list in column to genres
```

Out[4]:
```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'dat
e_added',
       'release_year', 'rating', 'duration', 'genre', 'description'],
      dtype='str')
```

In [5]:
```
df.drop(columns='show_id',inplace=True)  #--> Dropping the show_id column
```

In [6]:
```
df.duplicated().sum() #--> Checking for duplicates
```

Out[6]: np.int64(0)

In [7]:
```
df.isnull().sum() #--> Checking the missing values
```

Out[7]:
```
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
genre              0
description        0
dtype: int64
```

In [8]:
```
df.shape
```

Out[8]: (8807, 11)

In [9]:
```
df.dropna(subset=['date_added'],inplace=True)
df['date_added'].isna().sum()
```

Out[9]: np.int64(0)

In [10…
```
df.dropna(subset=['rating'],inplace=True)
df['rating'].isna().sum()
```

Out[10]: np.int64(0)

In [11…
```
df.dropna(subset=['duration'],inplace=True)
df['duration'].isna().sum()
```

Out[11]: np.int64(0)

In [12…
```
df.isnull().sum()
```

Out[12]:
```
type               0
title              0
director        2621
cast             825
country          829
date_added         0
release_year       0
rating             0
duration           0
genre              0
description        0
dtype: int64
```

In [13…
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 8790 entries, 0 to 8806
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   type          8790 non-null   str
 1   title         8790 non-null   str
 2   director      6169 non-null   str
 3   cast          7965 non-null   str
 4   country       7961 non-null   str
 5   date_added    8790 non-null   str
 6   release_year  8790 non-null   int64
 7   rating        8790 non-null   str
 8   duration      8790 non-null   str
 9   genre         8790 non-null   str
 10  description   8790 non-null   str
dtypes: int64(1), str(10)
memory usage: 3.8 MB
```

In [14…

```
df['date_added']=pd.to_datetime(df['date_added'],errors='coerce')
df['date_added'].isnull().sum()
```

Out[14]:

```
np.int64(88)
```

In [15…

```
df.dropna(subset=['date_added'],inplace=True)
df['date_added'].isna().sum()
```

Out[15]:

```
np.int64(0)
```

In [16…

```
df['day'] = df['date_added'].dt.day
df['month'] = df['date_added'].dt.month
df['year'] = df['date_added'].dt.year
df['month_name'] = df['date_added'].dt.month_name()
df['weekday'] = df['date_added'].dt.day_name()

df.head()
```

Out[16]:

	type	title	director	cast	country	date_added	release_year	rating	duration
0	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	PG-13	90 min
1	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons
2	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	TV-MA	1 Season
3	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	TV-MA	1 Season
4	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons

In [17…

```
df["duration"].unique()
```

Out[17]:

```
<ArrowStringArray>
[   '90 min', '2 Seasons',  '1 Season',    '91 min',   '125 min', '9 Se
asons',
   '104 min',   '127 min', '4 Seasons',    '67 min',
 ...
    '43 min',   '200 min',   '196 min',   '167 min',   '178 min',   '22
8 min',
    '18 min',   '205 min',   '201 min',   '191 min']
Length: 219, dtype: str
```

In [18…

```
s1='179 min'
s1.split()[0]
```

Out[18]:

```
'179'
```

In [19…

```
s1='12 Seasons'
int(s1.split()[0])
```

Out[19]: 12

In [20…
```
df["duration"]=df["duration"].apply(lambda x:int(x.split()[0]))
```

In [21…
```
df["duration"].unique()
```

Out[21]:
```
array([ 90,   2,   1,  91, 125,   9, 104, 127,   4,  67,  94,   5, 161,
        61, 166, 147, 103,  97, 106, 111,   3, 110, 105,  96, 124, 116,
        98,  23, 115, 122,  99,  88, 100,   6, 102,  93,  95,  85,  83,
       113,  13, 182,  48, 145,  87,  92,  80, 117, 128, 119, 143, 114,
       118, 108,  63, 121, 142, 154, 120,  82, 109, 101,  86, 229,  76,
        89, 156, 112, 107, 129, 135, 136, 165, 150, 133,  70,  84, 140,
        78,   7,  64,  59, 139,  69, 148, 189, 141, 130, 138,  81, 132,
        10, 123,  65,  68,  66,  62,  74, 131,  39,  46,  38,   8,  17,
       126, 155, 159, 137,  12, 273,  36,  34,  77,  60,  49,  58,  72,
       204, 212,  25,  73,  29,  47,  32,  35,  71, 149,  33,  15,  54,
       224, 162,  37,  75,  79,  55, 158, 164, 173, 181, 185,  21,  24,
        51, 151,  42,  22, 134, 177,  52,  14,  53,  57,  28,  50,  26,
        45, 171,  27,  44, 146,  20, 157, 203,  41,  30, 194, 233, 237,
       230, 195, 253, 152, 190, 160, 208, 180, 144, 174, 170, 192, 209,
       187, 172,  16, 186,  11, 193, 176,  56, 169,  40, 168, 312, 153,
       214,  31, 163,  19, 179,  43, 200, 196, 167, 178, 228,  18, 205,
       201, 191])
```

In [22…
```
df['rating'].unique()
```

Out[22]:
```
<ArrowStringArray>
[   'PG-13',    'TV-MA',       'PG',    'TV-14',    'TV-PG',     'TV-
Y',
    'TV-Y7',        'R',     'TV-G',        'G',    'NC-17',       'N
R',
 'TV-Y7-FV',       'UR']
Length: 14, dtype: str
```

Netflix has officially categorised these into 3 main categories

- Replacing `TV-Y`, `TV-Y7`, `G`, `TV-G`, `PG`, `TV-PG`, `TV-Y7-FV` to **Kids**
- `PG-13`, `TV-14` to **Teens**
- `R`, `TV-MA`, `NC-17` to **Adults**
- `NR`, `UR` to **null values**

In [23…
```
df['rating'] = df['rating'].replace(['TV-Y','TV-Y7','G','TV-G','PG','TV-P
df['rating'] = df['rating'].replace(['PG-13','TV-14'],'Teens')
df['rating'] = df['rating'].replace(['R','TV-MA','NC-17'],'Adults')
df['rating'] = df['rating'].replace(['NR','UR'],np.nan)
df.head()
```

Out[23]:

	type	title	director	cast	country	date_added	release_year	rating	duration
0	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	90
1	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	Adults	2
2	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	2021-09-24	2021	Adults	1
3	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	2021-09-24	2021	Adults	1
4	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	Adults	2

In [24…

```
df['rating'].unique()
```

Out[24]:

```
<ArrowStringArray>
['Teens', 'Adults', 'Kids', nan]
Length: 4, dtype: str
```

In [25…

```
df['rating'].isna().sum()
```

Out[25]:

```
np.int64(81)
```

In [26…

```
df.dropna(subset=['rating'],inplace=True)
df['rating'].isna().sum()
```

Out[26]:

```
np.int64(0)
```

In [27…

```
df.isna().sum()
```

Out[27]:
```
type               0
title              0
director        2535
cast             800
country          826
date_added         0
release_year       0
rating             0
duration           0
genre              0
description        0
day                0
month              0
year               0
month_name         0
weekday            0
dtype: int64
```

In [28…
```
df[(df['director'].isna()) & (df['country'].isna()) & (df['cast'].isna())
```

Out[28]:
```
96
```

In [29…
```
df.shape
```

Out[29]:
```
(8621, 16)
```

In [30…
```
df.dropna(subset=['director','country','cast'], how = 'all',inplace=True)
df.shape
```

Out[30]:
```
(8525, 16)
```

In [31…
```
df.isna().sum()
```

Out[31]:
```
type               0
title              0
director        2439
cast             704
country          730
date_added         0
release_year       0
rating             0
duration           0
genre              0
description        0
day                0
month              0
year               0
month_name         0
weekday            0
dtype: int64
```

In [32…
```
df["cast"][2].split(",")
```

Out[32]:
```
['Sami Bouajila',
 ' Tracy Gotoas',
 ' Samuel Jouy',
 ' Nabiha Akkari',
 ' Sofia Lesaffre',
 ' Salim Kechiouche',
 ' Noureddine Farihi',
 ' Geert Van Rampelberg',
 ' Bakary Diombera']
```

In [33…
```
df_cast = pd.DataFrame(df['cast'].apply(lambda x: str(x).split(', ')).tol
df_cast
```

Out[33]:

	0	1	2	3	4	5	6	
title								
Dick Johnson Is Dead	nan	NaN	NaN	NaN	NaN	NaN	NaN	
Blood & Water	Ama Qamata	Khosi Ngema	Gail Mabalane	Thabang Molaba	Dillon Windvogel	Natasha Thahane	Arno Greeff	Tsh
Ganglands	Sami Bouajila	Tracy Gotoas	Samuel Jouy	Nabiha Akkari	Sofia Lesaffre	Salim Kechiouche	Noureddine Farihi	G Ram
Kota Factory	Mayur More	Jitendra Kumar	Ranjan Raj	Alam Khan	Ahsaas Channa	Revathi Pillai	Urvi Singh	Aru
Midnight Mass	Kate Siegel	Zach Gilford	Hamish Linklater	Henry Thomas	Kristin Lehman	Samantha Sloyan	Igby Rigney	Ral
...	...	...	...	...	...	...	...	
Zinzana	Ali Suliman	Saleh Bakri	Yasa	Ali Al-Jabri	Mansoor Alfeeli	Ahd	NaN	
Zodiac	Mark Ruffalo	Jake Gyllenhaal	Robert Downey Jr.	Anthony Edwards	Brian Cox	Elias Koteas	Donal Logue	Joh
Zombieland	Jesse Eisenberg	Woody Harrelson	Emma Stone	Abigail Breslin	Amber Heard	Bill Murray	Derek Graf	
Zoom	Tim Allen	Courteney Cox	Chevy Chase	Kate Mara	Ryan Newman	Michael Cassidy	Spencer Breslin	
Zubaan	Vicky Kaushal	Sarah-Jane Dias	Raaghav Chanana	Manish Chaudhary	Meghna Malik	Malkeet Rauni	Anita Shabdish	Chit

8525 rows × 50 columns

In [34…
```
df_cast = df_cast.stack()
df_cast
```

Out[34]:
```
title
Dick Johnson Is Dead  0     nan
                      1     NaN
                      2     NaN
                      3     NaN
                      4     NaN
                           ...
Zubaan                45    NaN
                      46    NaN
                      47    NaN
                      48    NaN
                      49    NaN
Length: 426250, dtype: str
```

In [35…
```
df_cast = pd.DataFrame(df_cast)
df_cast
```

Out[35]:

		0
title		
Dick Johnson Is Dead	**0**	nan
	1	NaN
	2	NaN
	3	NaN
	4	NaN
...	**...**	...
Zubaan	**45**	NaN
	46	NaN
	47	NaN
	48	NaN
	49	NaN

426250 rows × 1 columns

In [36…
```
df_cast.reset_index(inplace=True)
df_cast
```

Out[36]:

	title	level_1	0
0	Dick Johnson Is Dead	0	nan
1	Dick Johnson Is Dead	1	NaN
2	Dick Johnson Is Dead	2	NaN
3	Dick Johnson Is Dead	3	NaN
4	Dick Johnson Is Dead	4	NaN
...	...	...	...
426245	Zubaan	45	NaN
426246	Zubaan	46	NaN
426247	Zubaan	47	NaN
426248	Zubaan	48	NaN
426249	Zubaan	49	NaN

426250 rows × 3 columns

In [37…

```
df_cast = df_cast[['title',0]]
df_cast
```

Out[37]:

	title	0
0	Dick Johnson Is Dead	nan
1	Dick Johnson Is Dead	NaN
2	Dick Johnson Is Dead	NaN
3	Dick Johnson Is Dead	NaN
4	Dick Johnson Is Dead	NaN
...	...	...
426245	Zubaan	NaN
426246	Zubaan	NaN
426247	Zubaan	NaN
426248	Zubaan	NaN
426249	Zubaan	NaN

426250 rows × 2 columns

In [38…

```
df_cast.columns=['title','cast']
df_cast
```

Out[38]:

	title	cast
0	Dick Johnson Is Dead	nan
1	Dick Johnson Is Dead	NaN
2	Dick Johnson Is Dead	NaN
3	Dick Johnson Is Dead	NaN
4	Dick Johnson Is Dead	NaN
...	...	...
426245	Zubaan	NaN
426246	Zubaan	NaN
426247	Zubaan	NaN
426248	Zubaan	NaN
426249	Zubaan	NaN

426250 rows × 2 columns

In [39…

```
df_cast.replace('nan',np.nan,inplace=True)
df_cast.isna().sum()
```

Out[39]:

```
title         0
cast     363310
dtype: int64
```

In [40…

```
df_genre = pd.DataFrame(df['genre'].apply(lambda x: str(x).split(', ')).t
df_genre = df_genre.stack()
df_genre = pd.DataFrame(df_genre)
df_genre.reset_index(inplace=True)
df_genre = df_genre[['title',0]]
df_genre.columns =['title','genre']
df_genre.replace('nan',np.nan,inplace=True)
df_genre.isna().sum()
```

Out[40]:

```
title       0
genre    6818
dtype: int64
```

In [41…

```
df.columns
```

Out[41]:

```
Index(['type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'genre', 'description', 'd
ay',
       'month', 'year', 'month_name', 'weekday'],
      dtype='str')
```

In [42…

```
df_director = pd.DataFrame(df['director'].apply(lambda x: str(x).split(',
df_director = df_director.stack()
df_director = pd.DataFrame(df_director)
df_director.reset_index(inplace=True)
```

```
df_director = df_director[['title',0]]
df_director.columns =['title','director']
df_director.replace('nan',np.nan,inplace=True)
df_director.isna().sum()
```

Out[42]:
```
title            0
director    103944
dtype: int64
```

In [43…
```
df_country = pd.DataFrame(df['country'].apply(lambda x: str(x).split(', '
df_country = df_country.stack()
df_country = pd.DataFrame(df_country)
df_country.reset_index(inplace=True)
df_country = df_country[['title',0]]
df_country.columns =['title','country']
df_country.replace('nan',np.nan,inplace=True)
df_country.isna().sum()
```

Out[43]:
```
title          0
country    92511
dtype: int64
```

In [44…
```
df_cast.drop_duplicates(inplace=True)
df_genre.drop_duplicates(inplace=True)
df_director.drop_duplicates(inplace=True)
df_country.drop_duplicates(inplace=True)
```

In [45…
```
df12=df_cast.merge(df_genre, on = 'title')
df12
```

Out[45]:

	title	cast	genre
0	Dick Johnson Is Dead	NaN	Documentaries
1	Dick Johnson Is Dead	NaN	NaN
2	Blood & Water	Ama Qamata	International TV Shows
3	Blood & Water	Ama Qamata	TV Dramas
4	Blood & Water	Ama Qamata	TV Mysteries
...	...	...	...
201906	Zubaan	Chittaranjan Tripathy	International Movies
201907	Zubaan	Chittaranjan Tripathy	Music & Musicals
201908	Zubaan	NaN	Dramas
201909	Zubaan	NaN	International Movies
201910	Zubaan	NaN	Music & Musicals

201911 rows × 3 columns

In [46…
```
df123=df_director.merge(df12, on = 'title')
df123
```

Out[46]:

	title	director	cast	genre
0	Dick Johnson Is Dead	Kirsten Johnson	NaN	Documentaries
1	Dick Johnson Is Dead	Kirsten Johnson	NaN	NaN
2	Dick Johnson Is Dead	NaN	NaN	Documentaries
3	Dick Johnson Is Dead	NaN	NaN	NaN
4	Blood & Water	NaN	Ama Qamata	International TV Shows
...	...	...	...	...
363722	Zubaan	NaN	Chittaranjan Tripathy	International Movies
363723	Zubaan	NaN	Chittaranjan Tripathy	Music & Musicals
363724	Zubaan	NaN	NaN	Dramas
363725	Zubaan	NaN	NaN	International Movies
363726	Zubaan	NaN	NaN	Music & Musicals

363727 rows × 4 columns

In [47…

```
df1234 = df_country.merge(df123, on = 'title')
df1234
```

Out[47]:

	title	country	director	cast	genre
0	Dick Johnson Is Dead	United States	Kirsten Johnson	NaN	Documentaries
1	Dick Johnson Is Dead	United States	Kirsten Johnson	NaN	NaN
2	Dick Johnson Is Dead	United States	NaN	NaN	Documentaries
3	Dick Johnson Is Dead	United States	NaN	NaN	NaN
4	Dick Johnson Is Dead	NaN	Kirsten Johnson	NaN	Documentaries
...	...	...	...	...	...
801966	Zubaan	NaN	NaN	Chittaranjan Tripathy	International Movies
801967	Zubaan	NaN	NaN	Chittaranjan Tripathy	Music & Musicals
801968	Zubaan	NaN	NaN	NaN	Dramas
801969	Zubaan	NaN	NaN	NaN	International Movies
801970	Zubaan	NaN	NaN	NaN	Music & Musicals

801971 rows × 5 columns

In [48…

```
df_new = df.merge(df1234,on="title")
df_new
```

Out[48]:

	type	title	director_x	cast_x	country_x	date_added	release_year	rating	du
0	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	
1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	
2	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	
3	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	
4	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	2021-09-25	2020	Teens	
...	...	...	...	...	...	...	...	...	
801966	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	Teens	
801967	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	Teens	
801968	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	Teens	
801969	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias,	India	2019-03-02	2015	Teens	

	type	title	director_x	cast_x	country_x	date_added	release_year	rating	du
				Raaghav Chanan...					
801970	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	2019-03-02	2015	Teens	

801971 rows × 20 columns

In [49…

```
df_new.drop(columns = ['cast_x','country_x','director_x','genre_x'],inpla
df_new
```

Out[49]:

	type	title	date_added	release_year	rating	duration	description	day	month
0	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9
1	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9
2	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9
3	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9
4	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9
...	...	...	...	...	...	...	...	...	...
801966	Movie	Zubaan	2019-03-02	2015	Teens	111	A scrappy but poor boy worms his way into a ty...	2	3
801967	Movie	Zubaan	2019-03-02	2015	Teens	111	A scrappy but poor boy worms his way into a ty...	2	3
801968	Movie	Zubaan	2019-03-02	2015	Teens	111	A scrappy but poor boy worms his way into a ty...	2	3
801969	Movie	Zubaan	2019-03-02	2015	Teens	111	A scrappy but poor boy worms his way into a ty...	2	3
801970	Movie	Zubaan	2019-03-02	2015	Teens	111	A scrappy but poor boy worms his way into a ty...	2	3

801971 rows × 16 columns

In [50…
```
df_new.rename(columns={'country_y':'country','director_y':'director','cas
```

In [51…
```
df_new.head()
```

Out[51]:

	type	title	date_added	release_year	rating	duration	description	day	month	year
0	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
1	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
2	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
3	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
4	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021

In [52…
```
df_new.shape
```

Out[52]:
```
(801971, 16)
```

In [53…
```
df_new['cast']=df_new['cast'].replace(np.nan,'Unknown')
df_new['country']=df_new['country'].replace(np.nan,'Unknown')
df_new['director']=df_new['director'].replace(np.nan,'Unknown')
```

In [54…
```
df_new.head()
```

Out[54]:

	type	title	date_added	release_year	rating	duration	description	day	month	year
0	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
1	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
2	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
3	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021
4	Movie	Dick Johnson Is Dead	2021-09-25	2020	Teens	90	As her father nears the end of his life, filmm...	25	9	2021

In [55…

```
df_new.drop(["description"],axis=1,inplace=True)
```

In [56…

```
df_new.duplicated().sum()
```

Out[56]:

```
np.int64(0)
```

In [57…

```
df_new.drop_duplicates(keep='first',inplace=True)
```

In [58…

```
df_movies=df_new[df_new['type']=='Movie']
df_tvs=df_new[df_new['type']=='TV Show']
```

In [59…

```
df_movies.shape
```

Out[59]:

```
(663837, 15)
```

In [60…

```
df_tvs.shape
```

Out[60]:

```
(138134, 15)
```

In [61…

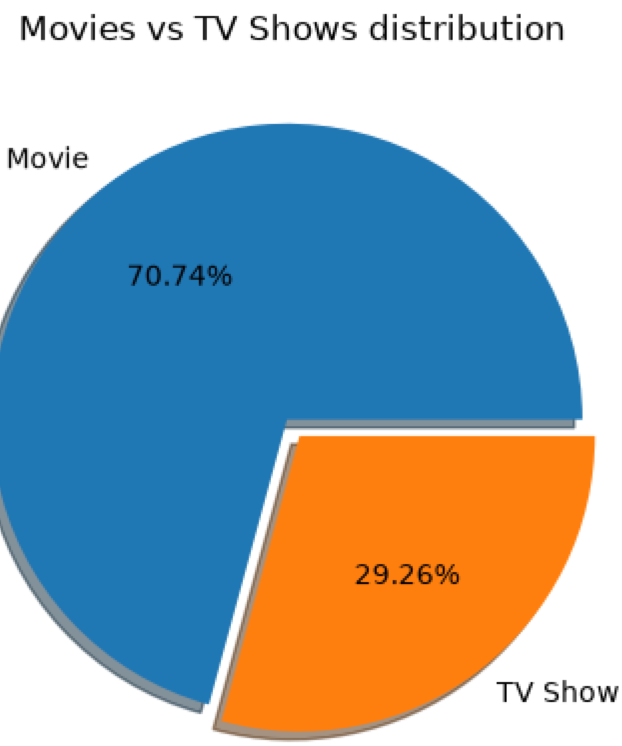
```
df_tvs.rename(columns={'duration':'seasons'},inplace=True)
```

⟠ Data Visualization

After completing the data cleaning and exploratory analysis, I started the **data visualization** phase to understand patterns, trends, and distributions in the Netflix dataset through different visualizations.

In [62…

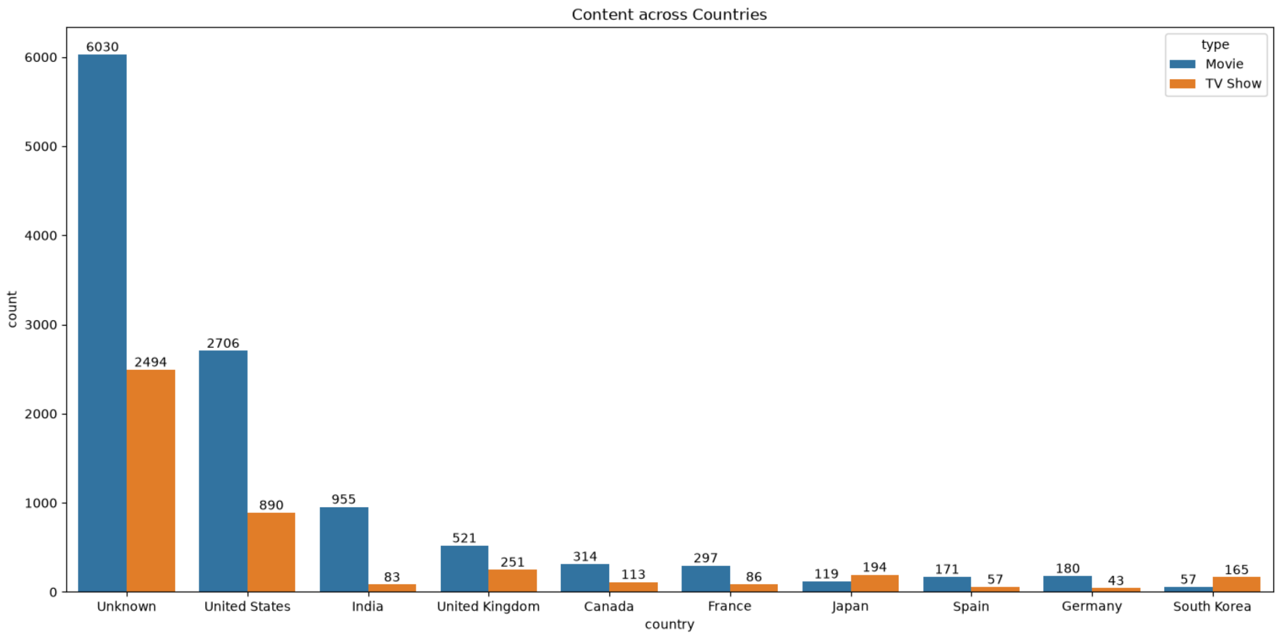
```
import matplotlib.pyplot as plt

plt.pie(df["type"].value_counts(), labels=df["type"].value_counts().index
        autopct="%1.2f%%", explode=[0.03,0.04], shadow=True)

plt.title("Movies vs TV Shows distribution")
plt.show()
```



In [63…

```
df_temp = df_new.drop_duplicates(subset=['country', 'title'])
df_temp

x = df_temp['country'].value_counts().head(10)
x

plt.figure(figsize=(17,8))
plt.title('Content across Countries')
label = sns.countplot(data=df_temp, x='country', hue='type', order=x.inde
for i in label.containers:
    label.bar_label(i)
```

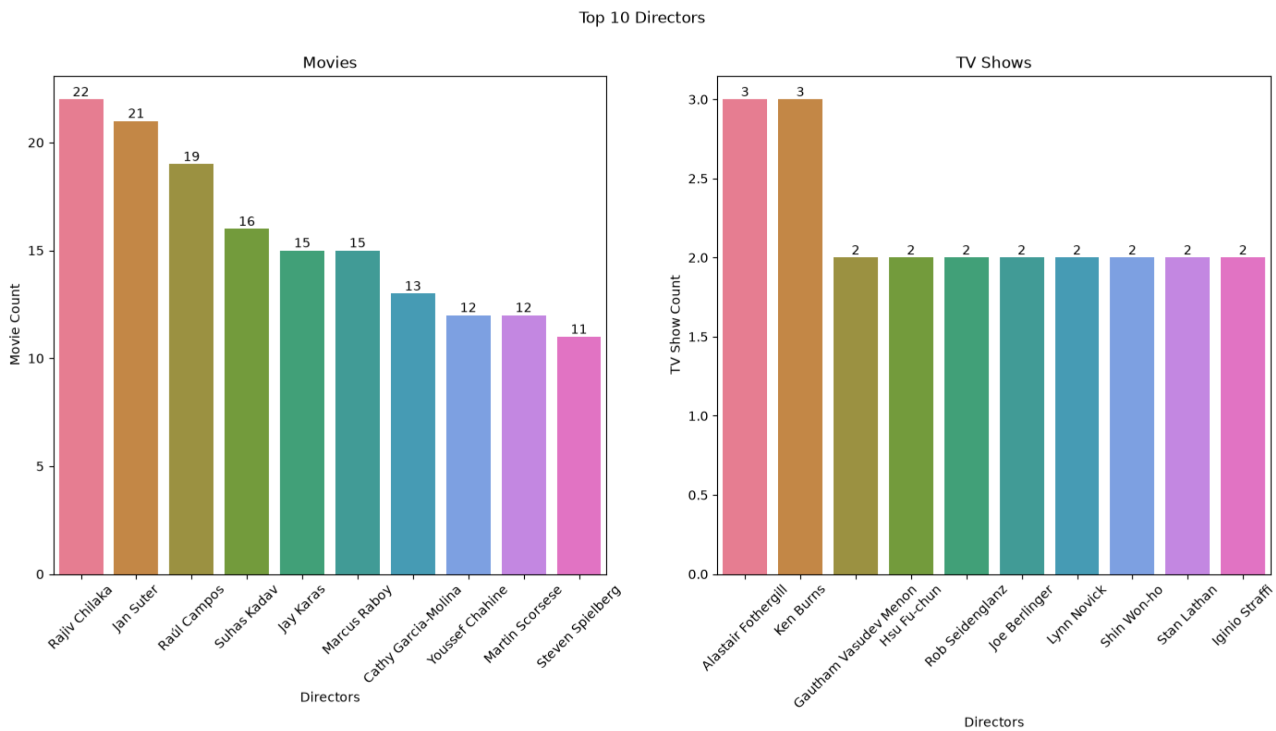
```
plt.show()
```



In [64…

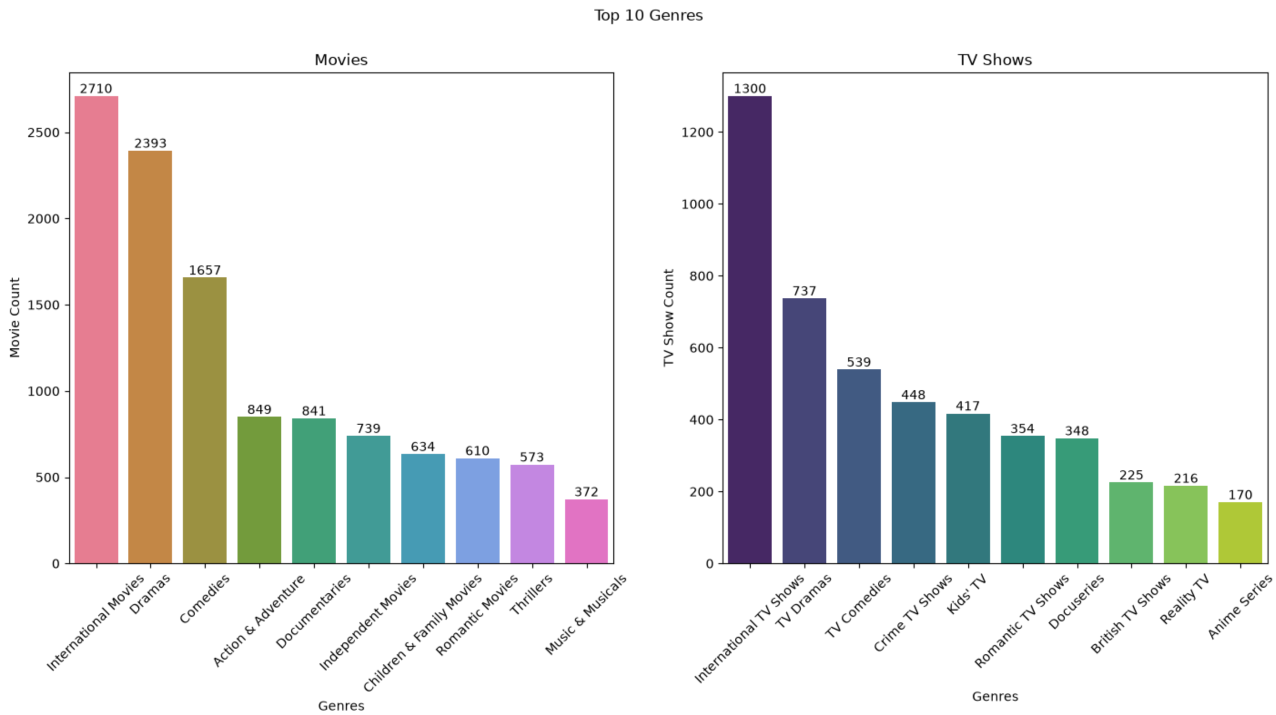
```
import warnings
warnings.filterwarnings('ignore')
df_movies_temp = df_movies.drop_duplicates(subset=['director', 'title'])
df_tv_temp = df_tvs.drop_duplicates(subset=['director', 'title'])

plt.figure(figsize=(17,7))
plt.suptitle('Top 10 Directors')

plt.subplot(1,2,1)
x = df_movies_temp['director'].value_counts()[1:11]
label = sns.countplot(data=df_movies_temp, x='director', order=x.index, p
for i in label.containers:
    label.bar_label(i)
plt.title('Movies')
plt.xticks(rotation=45)
plt.xlabel('Directors')
plt.ylabel('Movie Count')

plt.subplot(1,2,2)

y = df_tv_temp['director'].value_counts()[1:11]

label = sns.countplot(data=df_tv_temp, x='director', order=y.index, palet

for i in label.containers:
    label.bar_label(i)

plt.title('TV Shows')
plt.xticks(rotation=45)
plt.xlabel('Directors')
plt.ylabel('TV Show Count')
```

```
plt.show()
```



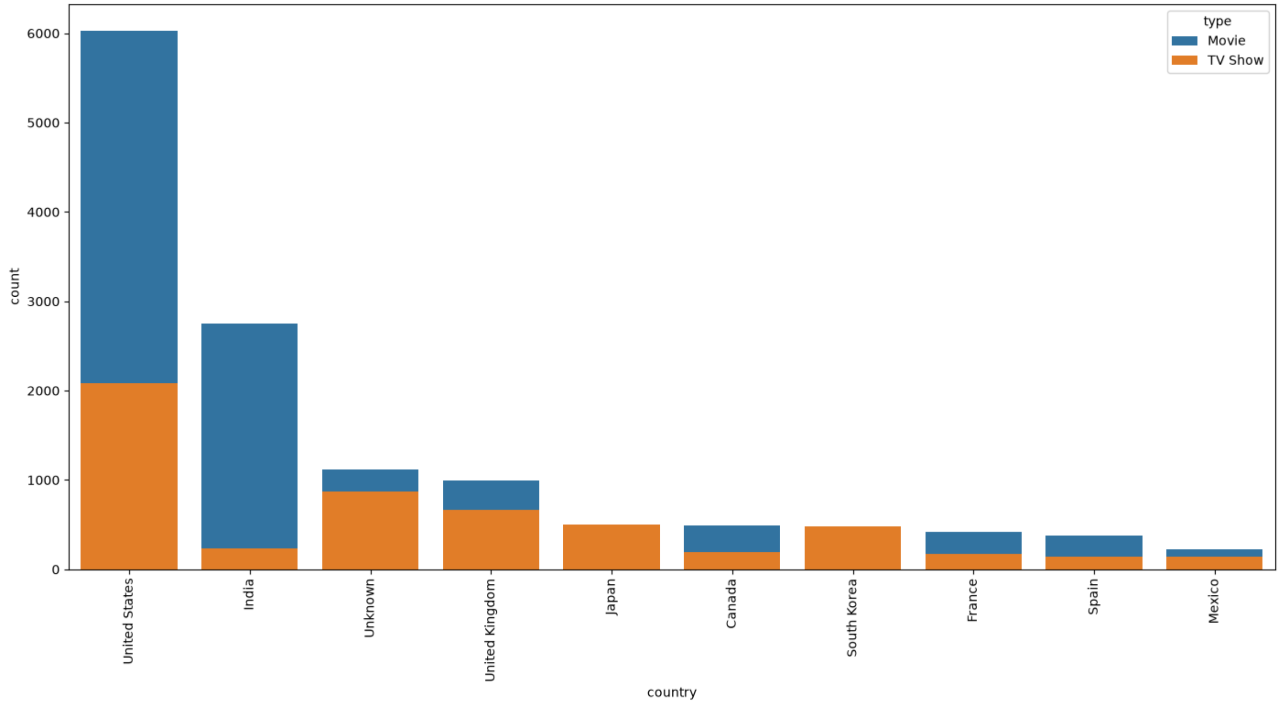
In [65…

```
df_movies_temp = df_movies.drop_duplicates(subset=['genre', 'title'])
df_tvs_temp = df_tvs.drop_duplicates(subset=['genre', 'title'])

plt.figure(figsize=(17,7))
plt.suptitle('Top 10 Genres')

plt.subplot(1,2,1)
x = df_movies_temp['genre'].value_counts().head(10)
label = sns.countplot(data=df_movies_temp, x='genre', order=x.index, pale
for i in label.containers:
    label.bar_label(i)
plt.title('Movies')
plt.xticks(rotation=45)
plt.xlabel('Genres')
plt.ylabel('Movie Count')

plt.subplot(1,2,2)
y = df_tvs_temp['genre'].value_counts().head(10)
label = sns.countplot(data=df_tvs_temp, x='genre', order=y.index, palette
for i in label.containers:
    label.bar_label(i)
plt.title('TV Shows')
plt.xticks(rotation=45)
plt.xlabel('Genres')
plt.ylabel('TV Show Count')

plt.show()
```



In [69…

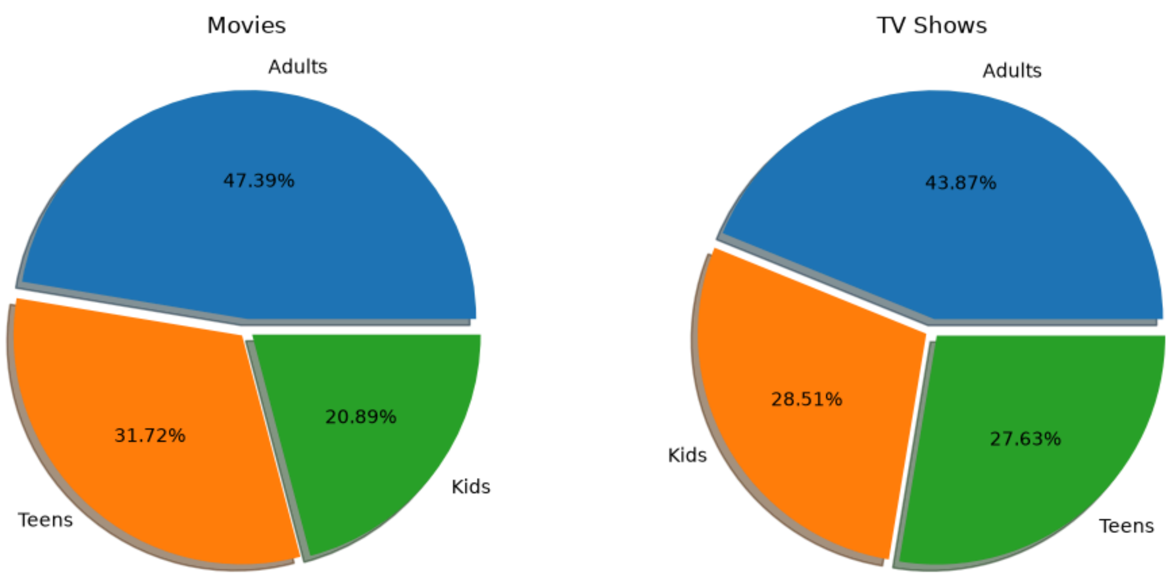
```
df_temp = df_new.drop_duplicates(subset=['genre', 'title'])

plt.figure(figsize=(17,8))
plt.xticks(rotation=90)

sns.countplot(
    data=df_temp,
    x='country',
    hue='type',
    dodge=False,
    order=df_temp['country'].value_counts().head(10).index
)

plt.show()
```



In [66…

```
df_movies_temp = df_movies.drop_duplicates(subset=['rating','title'])
df_tv_temp = df_tvs.drop_duplicates(subset=['rating','title'])

plt.figure(figsize = (12,6))
plt.suptitle('Classification of Content on Netflix')

plt.subplot(1,2,1)
plt.pie(df_movies_temp['rating'].value_counts(),
        labels = df_movies_temp['rating'].value_counts().index,
        autopct='%0.2f%%', explode = (0.05,0.03,0.03), shadow = True)
plt.title('Movies')

plt.subplot(1,2,2)
plt.pie(df_tv_temp['rating'].value_counts(),
        labels = df_tv_temp['rating'].value_counts().index,
        autopct='%0.2f%%', explode = (0.05,0.03,0.03), shadow = True)
plt.title('TV Shows')

plt.show()
```

Classification of Content on Netflix

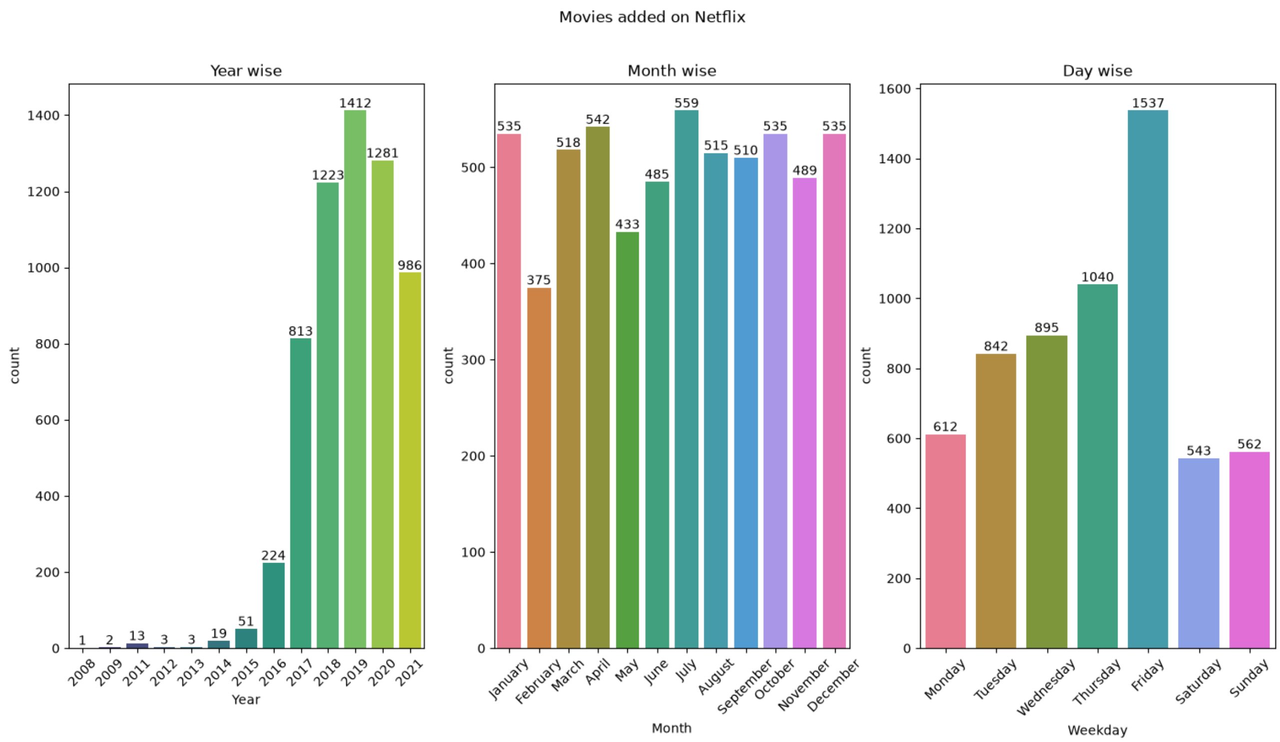


In [70…

```
df_movies_temp = df_movies.drop_duplicates(subset=['year', 'title'])
df_movies_temp


mv_year = df_movies_temp['year'].value_counts()
mv_year.sort_index(inplace=True)
mv_year


month_order = [
    'January',
    'February',
    'March',
    'April',
    'May',
    'June',
    'July',
    'August',
    'September',
    'October',
    'November',
    'December'
]

mv_month = df_movies_temp['month_name'].value_counts().loc[month_order]
mv_month


day_order = [
    'Monday',
    'Tuesday',
```

```
    'Wednesday',
    'Thursday',
    'Friday',
    'Saturday',
    'Sunday'
]

mv_day = df_movies_temp['weekday'].value_counts().loc[day_order]
mv_day


plt.figure(figsize=(17,8))
plt.suptitle('Movies added on Netflix')


plt.subplot(1,3,1)

label = sns.countplot(
    data=df_movies_temp,
    x='year',
    order=mv_year.index,
    palette='viridis'
)

for i in label.containers:
    label.bar_label(i)

plt.xticks(rotation=45)
plt.xlabel('Year')
plt.title('Year wise')


plt.subplot(1,3,2)

label = sns.countplot(
    data=df_movies_temp,
    x='month_name',
    order=mv_month.index,
    palette='husl'
)

for i in label.containers:
    label.bar_label(i)

plt.xticks(rotation=45)
plt.xlabel('Month')
plt.title('Month wise')


plt.subplot(1,3,3)

label = sns.countplot(
    data=df_movies_temp,
    x='weekday',
```

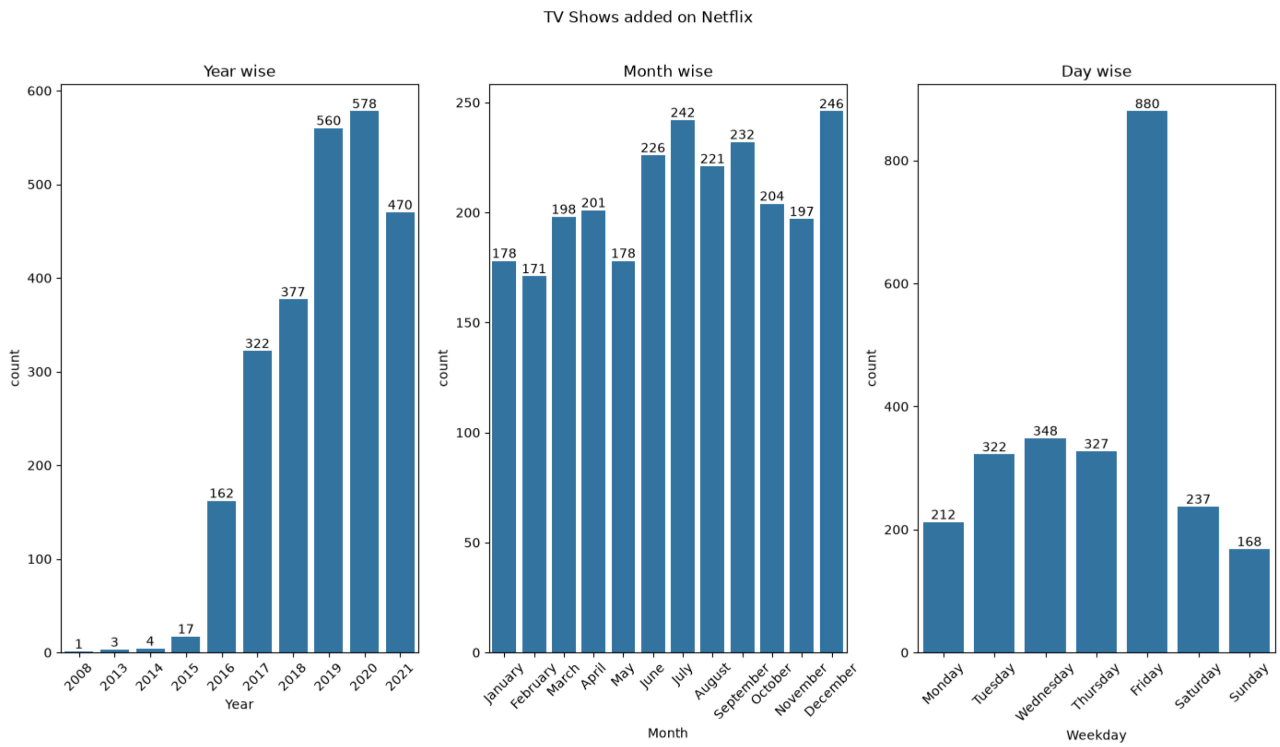
```
    order=mv_day.index,
    palette='husl'
)

for i in label.containers:
    label.bar_label(i)

plt.xticks(rotation=45)
plt.xlabel('Weekday')
plt.title('Day wise')


plt.show()
```



In [67…

```
df_tvs_temp = df_tvs.drop_duplicates(subset=['year','title'])

tv_year = df_tvs_temp['year'].value_counts()
tv_year.sort_index(inplace=True)

month_order = ['January', 'February', 'March', 'April', 'May', 'June', 'J
tv_month = df_tvs_temp['month_name'].value_counts().loc[month_order]

day_order = ['Monday','Tuesday','Wednesday','Thursday','Friday','Saturday
tv_day = df_tvs_temp['weekday'].value_counts().loc[day_order]


plt.figure(figsize=(17,8))
plt.suptitle('TV Shows added on Netflix')

plt.subplot(1,3,1)
```

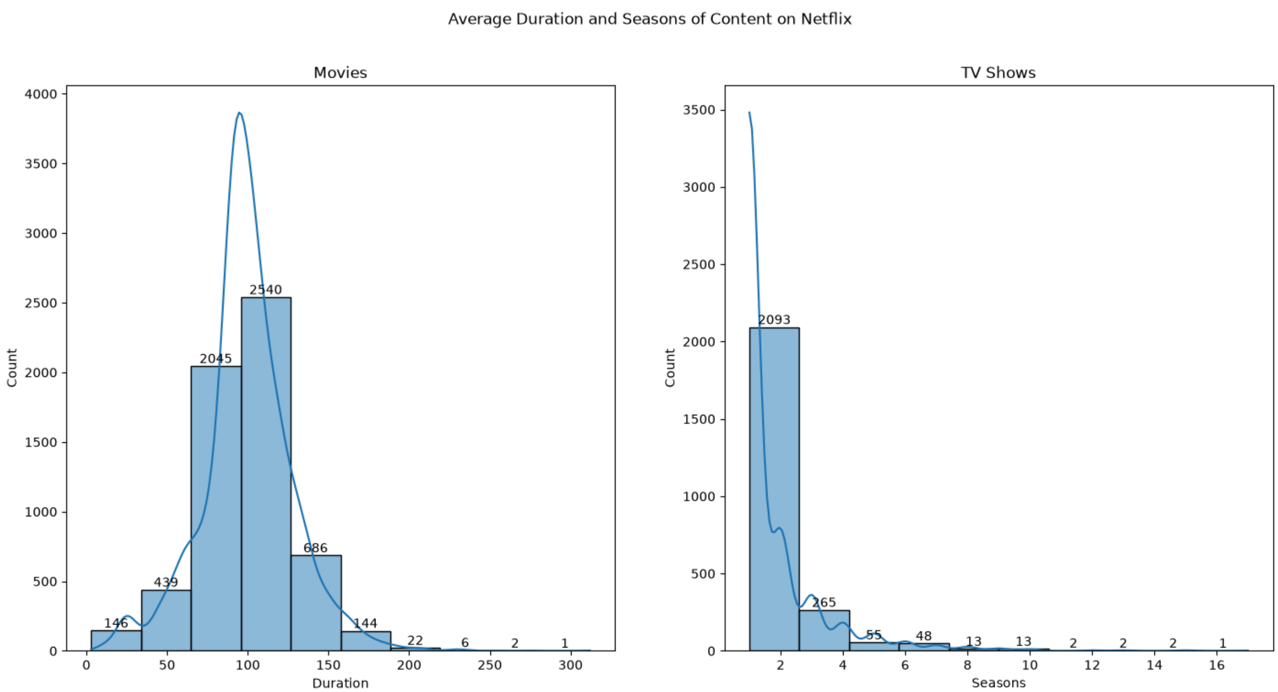
```python
label = sns.countplot(data=df_tvs_temp, x='year', order=tv_year.index)
for i in label.containers:
    label.bar_label(i)
plt.xticks(rotation=45)
plt.xlabel('Year')
plt.title('Year wise')

plt.subplot(1,3,2)
label = sns.countplot(data=df_tvs_temp, x='month_name', order=tv_month.in
for i in label.containers:
    label.bar_label(i)
plt.xticks(rotation=45)
plt.xlabel('Month')
plt.title('Month wise')

plt.subplot(1,3,3)
label = sns.countplot(data=df_tvs_temp, x='weekday', order=tv_day.index)
for i in label.containers:
    label.bar_label(i)
plt.xticks(rotation=45)
plt.xlabel('Weekday')
plt.title('Day wise')

plt.show()
```



In [68…

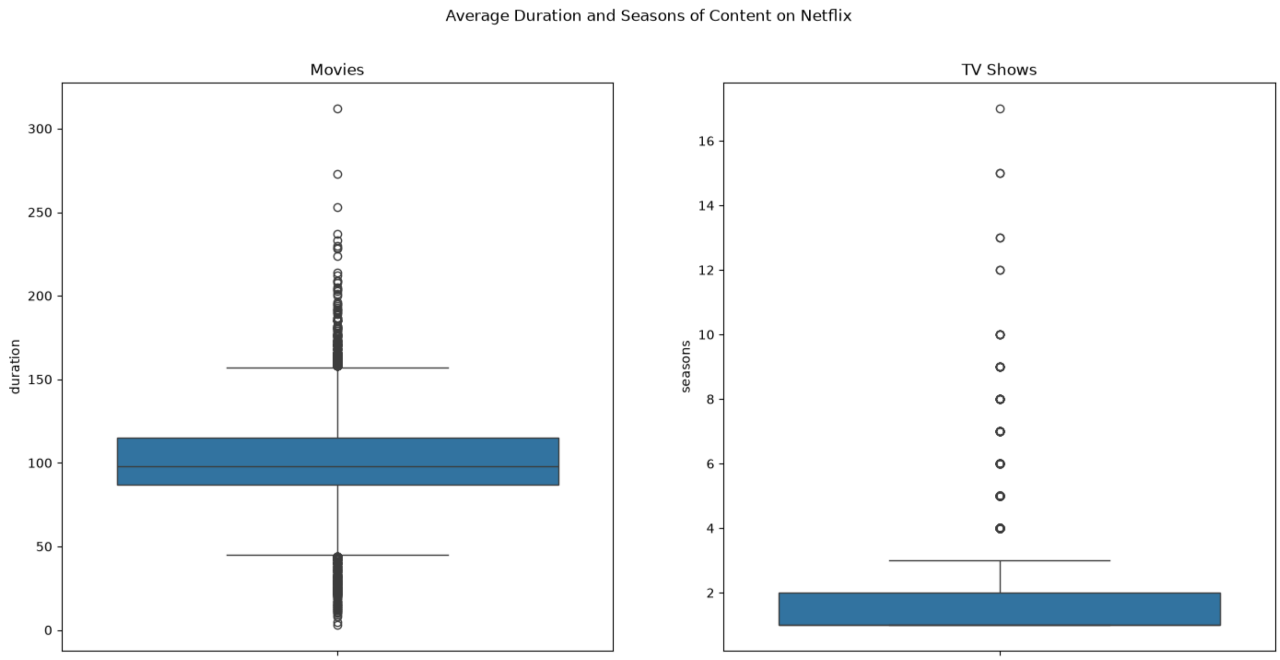
```python
df_movies_temp = df_movies.drop_duplicates(subset=['duration','title'])
df_tvs_temp = df_tvs.drop_duplicates(subset=['seasons','title'])

plt.figure(figsize=(17,8))
plt.suptitle('Average Duration and Seasons of Content on Netflix')
```

```
plt.subplot(1,2,1)
label = sns.histplot(df_movies_temp['duration'], bins=10, kde=True)
for i in label.containers:
    label.bar_label(i)
plt.xlabel('Duration')
plt.title('Movies')

plt.subplot(1,2,2)
label = sns.histplot(df_tvs_temp['seasons'], bins=10, kde=True)
for i in label.containers:
    label.bar_label(i)
plt.xlabel('Seasons')
plt.title('TV Shows')

plt.show()
```



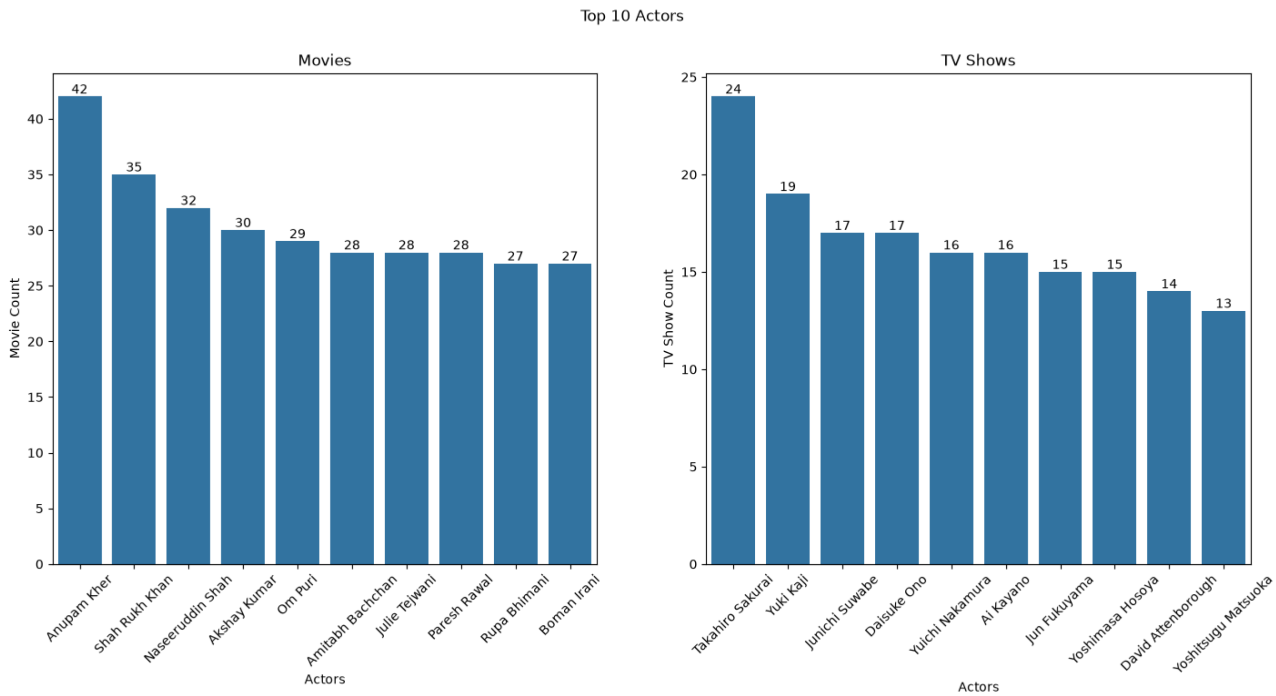
In [71…
```
plt.figure(figsize=(17,8))
plt.suptitle('Average Duration and Seasons of Content on Netflix')

plt.subplot(1,2,1)

sns.boxplot(
    y=df_movies_temp['duration']
)

plt.title('Movies')


plt.subplot(1,2,2)

sns.boxplot(
```

```
    y=df_tvs_temp['seasons']
)

plt.title('TV Shows')


plt.show()
```



In [73…

```
df_movies_temp = df_movies.drop_duplicates(subset=['cast','title'])
df_tvs_temp = df_tvs.drop_duplicates(subset=['cast','title'])

plt.figure(figsize=(17,7))
plt.suptitle('Top 10 Actors')

plt.subplot(1,2,1)
label = sns.countplot(data=df_movies_temp, x='cast', order=df_movies_temp
for i in label.containers: label.bar_label(i)
plt.title('Movies')
plt.xticks(rotation=45)
plt.xlabel('Actors')
plt.ylabel('Movie Count')

plt.subplot(1,2,2)
label = sns.countplot(data=df_tvs_temp, x='cast', order=df_tvs_temp['cast
for i in label.containers: label.bar_label(i)
plt.title('TV Shows')
plt.xticks(rotation=45)
plt.xlabel('Actors')
plt.ylabel('TV Show Count')

plt.show()
```



Insights from Data Visualization

- **Movies vs TV Shows:** Movies account for **70.74%** of the total content, while TV Shows account for **29.26%**, showing that Netflix has a much larger movie library.

- **Content Across Countries:** The **United States** has the highest amount of Netflix content among the identified countries, with movies contributing much more than TV Shows.

- **Country-wise Content:** India is the second-largest contributor, with a strong dominance of Movies, while countries such as Japan and South Korea have a comparatively higher contribution of TV Shows.

- **Top Directors – Movies: Rajiv Chilaka** is the leading director in the movie dataset with **22 movies**, followed by **Ian Suter (21)** and **Raul Campos (19)**.

- **Top Directors – TV Shows: Alastair Fothergill** and **Ken Burns** are among the leading TV Show directors, with **3 shows each**.

- **Top Movie Genres: International Movies** are the most common movie genre with **2710 titles**, followed by **Dramas (2393)** and **Comedies (1657)**.

- **Top TV Show Genres: International TV Shows** dominate with **1300 titles**, followed by **TV Dramas (737)** and **TV Comedies (539)**.
- **Content Classification – Movies:** Among movies, **Adults** category has the highest share at **47.39%**, followed by **Teens (31.72%)** and **Kids (20.89%)**.
- **Content Classification – TV Shows:** Among TV Shows, **Adults** represent **43.87%**, followed by **Kids (28.51%)** and **Teens (27.63%)**.
- **Movies Added by Year:** Netflix added the highest number of movies in **2019**, with **1412 movies**, followed by **2020 (1281)** and **2018 (1223)**.
- **Movies Added by Month: July** was the month with the highest number of movie additions, with **559 movies**, while **May** had the lowest with **433 movies**.
- **Movies Added by Weekday: Friday** had the highest number of movie additions with **1537**, showing that movie additions were strongly concentrated toward the end of the working week.
- **TV Shows Added by Year:** The highest number of TV Shows were added in **2020**, with **578 shows**, followed by **2019 (560)** and **2021 (470)**.
- **TV Shows Added by Month: December** had the highest number of TV Show additions with **246**, followed by **July (242)** and **September (232)**.
- **TV Shows Added by Weekday: Friday** had the highest number of TV Show additions with **880**, while **Sunday** had the lowest with **168**.
- **Movie Duration:** Most movies have a duration concentrated around **90–120 minutes**, with the distribution showing that standard-length movies are much more common than very short or very long movies.
- **TV Show Seasons:** Most TV Shows have **1–2 seasons**, with **1-season shows** being the most common. Shows with many seasons are comparatively rare.
- **Movie Duration Outliers:** The boxplot shows several outliers above the normal duration range, indicating that some movies have exceptionally long runtimes.
- **TV Show Season Outliers:** The boxplot shows several TV Shows with

more than **3 seasons**, with a few extending to **10+ seasons** and one reaching **17 seasons**.

- **Top Actors – Movies: Anupam Kher** appears most frequently among the movie actors with **42 movies**, followed by **Shah Rukh Khan (35)** and **Naseeruddin Shah (32)**.

- **Top Actors – TV Shows: Takhiro Sakurai** has the highest appearance count among TV Show actors with **24 shows**, followed by **Yuki Kaji (19)**.

- **Overall Insight:** The analysis shows that Netflix's content library is strongly dominated by **Movies**, with **International Movies, International TV Shows, Dramas and Comedies** being major genres. Content additions increased significantly during **2018–2020**, and **Friday** was the most common day for adding both Movies and TV Shows.

In [ ]:

```
```